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An improved quasi-steady model capable of calculating flexible deformation for bird-sized flapping wings

Tianyou MAO · Chuangqiang GUO · Bosong Duan

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Abstract Flapping wing air vehicles (FWAVs) are a type of bio-inspired aerial robot, whose application potential is currently receiving significant attention. However, the dynamic modeling techniques for flapping motion remain underdeveloped, particularly for bird-sized FWAVs. Traditional quasi-steady (QS) models' accuracy heavily relies on empirical coefficients, leading to unacceptable errors when analyzing flapping motion at different scales and wing flexibilities. Some improved reduced-order models also struggle to describe the effects caused by irregular flexible deformations of the wing surface. This paper presents an improved QS model that incorporates the interaction mechanism between wing flexible deformation and relative flow state, enabling quantitative analysis of wing deformation's aerodynamic influence and significantly reducing reliance on empirical coefficients. The model comprises two solvers: the dynamic solver that calculates the corresponding

aerodynamic loads based on the deformation state of the wing surface, and the structural solver that determines the new deformation state based on the load distribution on the wing surface. In each iteration cycle, these solvers exchange their output data until they converge to a stable wing surface displacement distribution and output the predicted aerodynamic data. Wind tunnel experiments, utilizing high-speed cameras and six-dimensional force sensors, validated the reliability of both the structural solver and the dynamic solver. This aerodynamic model, balancing efficiency and accuracy, shows strong potential for applications in FWAVs' structural and controller design, including but not limited to airfoil optimization and flight attitude stabilization.

Keywords Flapping wing air vehicles · Dynamic modeling · Aerodynamic · Wind tunnel experiment

T. MAO · C. GUO (✉)
State Key Laboratory of Robotics and System, Harbin
Institute of Technology, Harbin 150001, China
e-mail: chuanguang.guo@hit.edu.cn

C. GUO
HIT Wuhu Robot Technology Research Institute,
Wuhu 241008, China

B. Duan
School of Mechanical Engineering, Zhejiang University,
Hangzhou 310027, China

1 Introduction

Since the twenty-first century, the complex mechanisms of flapping flight have gradually been unveiled. With the continuous advancement of microelectromechanical systems (MEMS) and robotics technology, flapping flight has been realized on the scale of small unmanned aerial vehicles (UAVs). These novel bio-inspired aerial robots, known as FWAVs, have garnered significant attention for their potential

applications. However, the propulsion of bio-inspired FWAVs relies entirely on wing flapping, and the wing surfaces are flexible structures composed of various mechanical properties materials, resulting in highly complex flow fields during flapping flight. This complexity renders existing dynamic modeling techniques for flapping motion insufficiently mature [1, 2], often necessitating extensive wind tunnel experiments to determine the effectiveness of FWAV structures and control strategies, thereby impeding their development.

The flight mechanisms of flapping motion differ significantly from those of traditional propeller-based mechanisms, with propulsion generated by wing flapping at low Reynolds numbers [3, 4], characterized by high temporal variability, nonlinearity, and difficulty in quantification [5]. Numerous factors influence the aerodynamic performance of flapping motion, including flapping frequency, amplitude, wing mass and stiffness distribution, body motion posture, and flight speed [6–9]. Steady-state models based on momentum or actuator disc theory, lifting line theory, or vortex ring theory struggle to accurately describe such complex and variable motion [2, 10, 11]. The unsteady model typically assumes that the starting point of the wake satisfies the Kutta-Zhukovsky condition, which raises doubts about its applicability in handling the repeated bidirectional rotation in flapping-wing motion [2, 12]. While steady-state and unsteady models may be more effective under certain conditions, the QS model is still the most commonly used method for flapping wing dynamics modeling at this stage.

The core idea of the QS model is to decompose the unsteady flow problem into a series of instantaneous steady-state flow problems [13]. At each moment, the flow is assumed to be stable. The model is established by discretizing the flapping process along the time axis, treating the continuous wing surface changes as a series of static phases, and analyzing the aerodynamic characteristics of each phase under steady-state conditions. Chin et al. [14] summarized the aerodynamic mechanisms in flapping motion as: (1) leading-edge vortices (LEV), (2) Wagner effect, (3) clap and fling, (4) rotational circulation, (5) wing-wake interaction, and (6) added mass. Based on these aerodynamic mechanisms, Dickinson et al. [15–17] studied fruit fly experimental data and proposed a QS model based on empirical coefficients, considering translational

circulation, rotational circulation, wake capture effects, and added mass effect. This QS model, currently the most widely applied, eliminates the need to solve the complex process of unsteady flow, enabling rapid prediction of instantaneous aerodynamics in flapping motion. However, its accuracy is entirely dependent on empirical coefficients [2].

Empirical coefficients derived from experimental methods struggle to accommodate diverse structural and flight parameters. The translational circulation coefficient in Dickinson's model is expressed as a single-variable function of the angle of attack (AOA) [16], failing to account for factors like wing vein layout, wing shape, and wing material. Han et al. [18] improved the expression of the translational circulation coefficient based on the variation of the aerodynamic center with AOA, but the revised empirical coefficients still cannot be universally applied to all types of FWAVs. This empirical data-driven formula remains accurate for micro-scale FWAVs due to typically low disturbance indoor testing environments and flight modes primarily involving hovering or low-speed translation [19–22]. However, for bird-sized FWAVs performing high-speed operations in outdoor environments with wind disturbances [23–25], flexible wing deformation significantly alters local flow states. Constructing an aerodynamic model with a wide range of uncertain flight parameters through empirical coefficients is challenging, requiring extensive experimental data and incurring high costs in terms of time and money.

Investigating the impact of flexible wing deformation on aerodynamic characteristics is one of research topics for advanced FWAVs [26]. Heathcote et al. [27] noted in their experiments that flexible wing deformation slightly increases thrust but reduces lift for FWAVs. Pourtakdoust et al. [28] studied the propulsion mechanism of FWAVs in a no-wind, zero-degree absolute AOA environment and developed an aeroelastic model, concluding that active propulsion for bird-sized FWAVs is entirely driven by wing flexible deformation. Liu et al. [29] used computational fluid dynamics (CFD) methods to establish a two-dimensional motion simulation model of flexible wings compared to rigid wings, finding that flexible wings significantly enhance FWAV thrust. The research team from the University of Florida [30, 31] conducted extensive studies on wing flexibility and observed that significant wing torsion generates noticeable vortices

and high-speed airflow. This indicates that softer wings can create a stronger velocity field around the wing, thereby producing greater thrust. Qin et al. [32] analyzed the deformation process of flexible wings and wing veins using the Workbench platform, indicating that increased wing flexibility results in increased drag. Quantifying the influence of flexible wing deformation on aerodynamics might be addressed using fluid–structure interaction (FSI) based CFD methods. However, solving the overall FSI equations involves a high-dimensional non-homogeneous system with issues of ill-conditioning and computational intensity [7, 33]. Many improved FSI solvers for FWAVs [34–37] ensure numerical stability but remain computationally expensive and require substantial fluid dynamics knowledge for solver creation, limiting widespread application.

Rapidly and accurately describing the distribution of flexible wing deformation and its aerodynamic interactions is a critical challenge. To measure real-time flexible wing deformation, Yuan et al. [38] used depth cameras for dynamic 3D reconstruction of FWAVs. Tay et al. [39] proposed an experimental and data-driven hybrid method, capturing wing deformation during flapping with high-speed cameras and using it as input for numerical solvers, bypassing complex and resource-intensive FSI problems. While effective for micro-scale FWAVs, these methods may not suit bird-sized FWAVs in more complex flight environments. Nakata et al. [40] fitted CFD simulation data using least squares and proposed a CFD-informed QS model, validated with hovering hawkmoths, outperforming traditional QS model. It is undeniable that once a data-driven model is successfully established, it can achieve both high accuracy and rapidity, making it a potentially optimal approach for real-time control. However, constructing the database for this method requires significant computational resources, especially when the wing exhibits high flexibility. To ensure accuracy, bidirectional fluid–structure interaction (FSI) must be employed, further increasing the computational demand.

Developing reduced-order models (ROMs) tailored for flapping-wing motion might provide a good compromise between computational efficiency and accuracy [41, 42]. Some existing research focuses on enhancing the aerodynamic empirical coefficients [43–45] in QS models or refining the approximation of unsteady terms [46, 47]. These approaches are highly

effective when the wing flexibility is moderate or low, and their rapidity can even support real-time control [24, 48]. However, for bird-sized flapping-wing vehicles, the wing's torsion is completely governed by irregular, unpredictable passive deformations, making it challenging to quantify the relationship between flexible deformation and aerodynamic characteristics using one or several coefficients. A potentially effective solution may be to analyze the flexible deformation and aerodynamic characteristics separately, then integrate them into a comprehensive model. Schwab et al. [49] developed a quasi three-dimensional reduced-order model based on Deformable Blade Element Theory (DBET), capable of describing the thrust and deformation profile of flexible wings. Singh and Chopra [50] developed a structural model based on the homogenized plate finite element method to quantify the wing's flexible deformation and its aerodynamic effects. However, these methods may not be suitable for wings with highly uneven stiffness distributions, and they may struggle to intuitively reflect how flexible deformation influences aerodynamic characteristics.

In this context, our study proposes an improved QS model that considers the interaction mechanism between wing flexible deformation and relative flow state. It consists of two modules: dynamic solver and structural solver. The QS model in the dynamic solver incorporates translational circulation, rotational circulation, added mass, inertial forces, and unsteady terms. Meanwhile, the finite element model specifically designed for bird-sized FWAVs is developed in the structural solver. An implicit step-by-step algorithm is used to iteratively update certain terms in the QS model, enabling a clear and quantitative description of how flexible deformation affects the aerodynamic forces. This improved QS model requires 3 to 4 orders of magnitude less time to compute a set of parameter combinations compared to conventional CFD methods, with accuracy validated through experiments using high-speed cameras and six-dimensional force sensors. This modeling method, which effectively balances computational efficiency and accuracy, demonstrates strong potential for application in structural and controller design.

The rest of the article is organized as follows. Section II introduces the modeling procedure, including the principles of the dynamic solver, structural solver, and partitioned implicit algorithm. Section III details the wing flexible deformation and aerodynamic

characteristics monitoring experiments conducted in a wind tunnel environment to verify the reliability of the modeling method. Section IV designs two kinds of FWAVs prototypes based on the proposed theory and verifies their flight capabilities. Section V summarizes the results and outlines future work.

2 Flexible deformation-relative flow state interaction modeling

In the existing research on flapping wing aerodynamics, it is common to combine the QS model with the blade element model (BEM) [40, 51, 52]. This approach significantly simplifies the modeling process but introduces the issue of solution accuracy being heavily dependent on empirical coefficients [2]. This problem is particularly pronounced for bird-sized FWAVs with larger wing sizes and higher flight speeds, where significant flexible deformation causes notable changes in relative flow conditions. These effects are challenging to describe accurately using empirical coefficients alone. In this section, we design a dynamic solver based on the QS model and a structural solver based on the finite element model (FEM). The interaction between wing flexibility and relative flow conditions is then analyzed using a partitioned implicit algorithm.

2.1 Dynamic solver

First, Fig. 1 provides a clear definition of the coordinate systems and angular quantities involved in this study. In Fig. 1a, the Z-axis of all coordinate systems can be confirmed by the right-hand rule. The body coordinate system is defined as $O_bX_bY_bZ_b$, where O_b coincides with the center of mass of the aircraft, X_b points to the front of the aircraft, and Y_b points to the right of the aircraft. The trajectory coordinate system is defined as $O_tX_tY_tZ_t$, where X_t always aligns with the direction of the aircraft's velocity, and Y_t coincides with Y_b . The wing coordinate system is defined as $O_wX_wY_wZ_w$, where X_w points to the tail of the aircraft and Y_w aligns with the midline of the wing's leading edge. The absolute angle of attack α_0 is the angle between X_t and X_b , while the flapping angle θ is the angle between Y_w and Y_b . In Fig. 1b–d, the purple area represents the position of the wing in an ideal rigid state, the green area represents the flexible

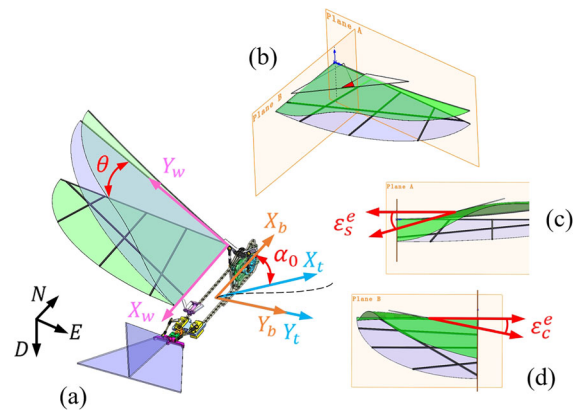


Fig. 1 Definitions of coordinate systems and angles. **a** The wing coordinate system, body coordinate system, trajectory coordinate system, and definition of the flapping angle. **b** An example illustrating how an infinitesimal plane formed by any three nodes on the wing surface can quantitatively describe the local characteristics of irregular flexible deformation. **c** Definition of the spanwise torsion angle of a wing section. **d** Definition of the chordwise torsion angle of a wing section

deformation state, and the red triangular area represents a differential element formed by any three nodes on the wing surface. The chordwise twist angle ϵ_c^e is defined as the complementary angle of the acute angle between the plane of the differential element and the $Y_wO_wZ_w$ plane, while the spanwise twist angle ϵ_s^e is the complementary angle of the acute angle between the plane of the differential element and the $X_wO_wZ_w$ plane. The positive directions for both angles are illustrated in Fig. 1c and d.

Since our study focuses on bird-sized flapping wings, which have relatively large wing areas and significant flexibility, their flexible deformation is not reflected as a uniform twisting of the entire wing. Instead, different parts of the wing surface exhibit irregular shapes of bulges or depressions. Therefore, unlike many studies on insect-scale wings, where the twisting motion can be approximated with kinematic equations [5, 31, 43, 46], this approach is not applicable here, meaning we cannot solve the problem using BEM. To more accurately characterize the relative flow state, this study uses two-dimensional FEM to divide the wing surface into small meshes. In this section, we will introduce the method of solving loads in the finite element mesh.

First, define U_∞ as the free-stream velocity and U_t^e as the characteristic velocity of the flapping motion. U_t^e can be expressed as the vector sum of the free-

stream velocity and the flapping velocity at this differential element [17, 53], as follows:

$$U_t^e = U_\infty + \omega_\lambda \times r_s^e \quad (1)$$

where r_s^e is the spanwise distance from the geometric center of the elemental area to the wing's rotation axis in the wing coordinate system. The flapping angular velocity vector ω_λ of the flapping motion can be expressed as the time derivative of the flapping angle λ , with its negative direction defined as coinciding with the X_w -axis. The flapping angle of the wing can be represented as:

$$\lambda = \frac{\Lambda}{2} g(f, t) \quad (2)$$

where Λ is the absolute value of the difference between the upper and lower limit angles of a flapping stroke, and $g(f, t)$ represents the periodic function describing the flapping motion gait. Due to mechanical constraints in generating flapping motion for bird-sized FWAVs, $g(f, t)$ is typically a sinusoidal function [54, 55], which can be expressed as $g(f, t) = \sin(2\pi ft)$.

When establishing the dynamic model, a semi-empirical approach was adopted. The aerodynamic loads were obtained by superimposing quasi-steady translational, rotational, and added mass effects. This study also referenced the reduced-order model proposed by Taha et al. [46], incorporating the effects of unsteady phenomena. The instantaneous dynamic equation at each discrete time point can be expressed as [48]:

$$F_{inst} = F_{tran} + F_{rot} + F_{add} + F_{uns} + F_{iner} \quad (3)$$

where F_{tran} is the translational circulatory component, F_{rot} is the rotational circulatory component, F_{add} is the added mass force, F_{uns} represents the unsteady effects, and F_{iner} is the inertial force. The circulatory components were calculated based on the commonly used Dickinson model [17], but not established within the traditional BEM framework. The expression for the translational circulatory force based on the two-dimensional finite element mesh can be represented as [17]:

$$F_{tran} = \frac{1}{2} \rho \int_A C_{tran}^e U_t^{e2} ds^e \quad (4)$$

where ρ is the fluid density, S^e is the effective

projected area of the differential element, and C_{tran}^e is the translational circulatory coefficient at the differential element, which is a function of α^e , the relative angle of attack. $C_{L,tran}$ and $C_{D,tran}$ represent the translational lift and drag coefficients, respectively. Since the FEM includes an additional dimension along the chordwise direction of the wing compared to BEM, it presents a challenging problem of describing the distribution of translational circulation forces along the chord. Precise modeling is evidently difficult. For the bird-sized FWAVs with moderate aspect ratios and moderate wing stiffness examined in this study, assuming a uniform distribution of translational circulation forces along the chord is an appropriate simplification. This significantly reduces the modeling difficulty without compromising the model's accuracy. Therefore, this simplification is adopted in the current modeling approach, assuming that the translational circulation coefficient is independent of the chordwise position. However, it is important to note that this simplification may not be suitable for some FWAVs with insufficient pretension [56] or butterfly-inspired FWAVs with low aspect ratios [57]. The corresponding values are based on the general model, which thoroughly considers the wake capture effect, proposed by Nabawy et al. [43]:

$$\begin{cases} C_{L,tran}^e = \left(\frac{0.5C_{Lx,2d}}{E + \frac{\pi AR}{kC_{Lx,2d}}} \right) \sin 2\alpha^e \\ C_{D,tran}^e = C_{L,tran}^e \tan \alpha^e \end{cases} \quad (5)$$

The coefficient $C_{Lx,2d}$ can be set to 0.09 deg⁻¹ (approximately 5.157 rad⁻¹) based on Okamoto et al.'s research [58]. The parameter E represents the edge correction factor in the lifting line theory, as proposed by Jones et al. [59], and is defined as the ratio of the wing's semiperimeter to the span. AR is the aspect ratio of the individual wing, and k accounts for the efficiency difference between the assumed ideal uniform downwash distribution and the actual downwash distribution. Based on the work of Nabawy et al. [60], k is defined as 1.5 for our bird-sized FWAVs.

The rotational circulation force arises due to the passive chordwise twist of the wing membrane, particularly significant when the wing membrane area is large and highly flexible [61, 62]. It can be expressed as [43]:

$$F_{rot} = \rho C_{rot} \int_A U_t^e \varepsilon_c^e r_c^e ds^e \quad (6)$$

$$C_{rot} = \pi(0.75 - \hat{x}_0) \quad (7)$$

where C_{rot} is the rotational circulatory coefficient, r_c^e is the chordwise distance from the geometric center of the elemental area to the wing's rotation axis in the wing coordinate system, $\hat{x}_0$ is the dimensionless position of the flapping motion's pitch axis relative to the leading-edge axis.

The added mass force can be expressed as [43]:

$$F_{add} = \frac{\pi}{4} \rho \int_A U_n^e r_c^e ds \quad (8)$$

where U_n^e represents the normal velocity component of the reference speed U_t^e relative to the air. According to the reduced-order model proposed by Taha et al. [46], the effects of unsteady phenomena can be approximated as:

$$F_{uns} = \frac{\pi}{4} \rho \int_A (\omega_\lambda \sin \alpha^e + \omega_\lambda \alpha^e \cos \alpha^e) \cos \alpha^e r_s^e r_c^e ds \quad (9)$$

According to Newton's second law, the inertial force can be expressed as:

$$F_m = \omega_\lambda \int_A \rho_w^e B_w^e r_s^e ds^e \quad (10)$$

where ρ_w^e and B_w^e are the density and the thickness of the wing membrane material.

Through coordinate transformation, the thrust force F_T , lift force F_L , and yaw centripetal force F_C can be obtained:

$$\begin{bmatrix} F_T \\ -F_L \\ F_C \end{bmatrix} = R_y(\alpha_0) R_x(\theta) \begin{bmatrix} 0 \\ 0 \\ F_{inst} \end{bmatrix} \quad (11)$$

where $R_i(\cdot) \in SO(3)$ represents the rotation matrix around the i axis.

2.2 Structural solver

The wing structure of biomimetic FWAVs mimics that of birds and has been simplified by reducing the complex structures of bones, muscles, and feathers to a planar wing network composed of carbon fiber wing

veins and elastic wing membranes. This simplification not only reduces the degrees of freedom in motion but also makes the stiffness distribution of the wing surface more traceable. However, accurately establishing the overall stiffness equation remains challenging due to several factors: (1) The very thin wing membrane lacks bending stiffness and exhibits nonlinear large deformations; (2) There is a complex coupling relationship between the in-plane stress of the wing membrane and the deformation of the wing veins.

Creating a high-precision, nonlinear three-dimensional finite element model is evidently too complex and computationally inefficient, making appropriate simplifications necessary. In fact, for typical bird-sized FWAVs, considering factors such as wear resistance and material costs, the wing membrane material usually does not have a very high elastic modulus [63, 64]. This implies that the flexible deformation of the wing membrane is continuously and smoothly distributed between the wing veins, allowing the deformation of the entire wing to be described by the deformation of a few key wing vein nodes. The following assumptions are made for the finite element solution process:

Assumption 1: The displacement of the wing membrane element nodes along the Z_w axis of the wing coordinate system is smoothly distributed between adjacent wing vein element nodes. The transverse displacement in the wing coordinate system is neglected, and the tensile stress of the membrane element is generated entirely by normal displacement-induced deformation.

Assumption 2: The wing vein elements only bear forces along the Z_w axis, including their own inertial forces and the Z_w component of the contact force with the membrane. The contact force at the boundary of the wing veins satisfies a linear distribution relationship.

It should be noted that these two main assumptions are only applicable when the wing membrane has an appropriate pre-tension (which applies to most bird-sized or insect-scale FWAVs). For some FWAVs with insufficient pre-tension (such as the prototype by Nan et al. [56]), similar assumptions cannot be made in the modeling process.

Based on these assumptions, the flexible deformation of the wing membrane can be roughly estimated by the distribution function between key wing vein

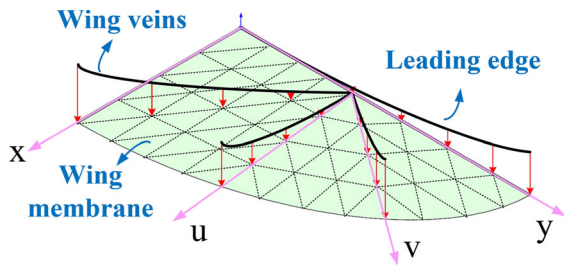


Fig. 2 Example of finite element mesh layout for the wing. xy represents the position coordinates in the wing coordinate system. uv refers to the generalized coordinates defined for the convenience of finite element modeling, with an example provided only for the corresponding wing surface area

nodes without precise calculation. This approach reduces the structural analysis to balancing aerodynamic loads and tensile stresses on the wing vein boundaries. The wing veins, bearing forces in one direction, can be modeled as elastic beams. Figure 2 shows the finite element mesh layout for a common wing shape, where three-node constant strain triangle (CST) elements [65] are used for the membrane and two-node elastic beam elements [66] for the wing veins. This meshing method ensures that wing vein and membrane nodes coincide, facilitating the application of generalized coordinates and boundary conditions.

For the membrane part, the external load on the i -th membrane element, $f_{i,inst}^e$, is considered to act on the center point of the element. The external load at the j -th node of the membrane element is expressed as:

$$P_{w,j}^{ext} = \sum \frac{f_{i,inst}^e}{3}, i = \{k\} \quad (12)$$

where the subscript w denotes the wing membrane element, $\{k\}$ denotes all membrane element numbers containing the j -th node.

The generalized coordinates $r = [uv]^T$ describe each membrane region segmented by the wing veins, with the wing vein nodes distributed on the coordinate axes and endpoints $(u_L, 0)$ and $(0, v_L)$. The coordinates of the j -th node of the membrane element are (u_j, v_j) , and the coordinate transformation is given by:

$$r_j(u_j, v_j) = J_{pr} p_j(x_j, y_j) \quad (13)$$

where $p_j(x_j, y_j)$ is the horizontal and vertical coordinates of the node in the wing coordinate system, and J_{pr} is the Jacobian matrix that maps p to r .

According to Assumption 1, if the normal displacement of the j -th wing membrane element node is h_j , then its in-plane coordinate change is:

$$\begin{cases} u_j' = \left(1 + \frac{h_j^2}{u_j^2 + v_j^2}\right)^{\frac{1}{2}} u_j \\ v_j' = \left(1 + \frac{h_j^2}{u_j^2 + v_j^2}\right)^{\frac{1}{2}} v_j \end{cases} \quad (14)$$

For a three-node CST element, the shape function matrix can be written as follows [65]:

$$N_i = \frac{1}{2A} (a_i + b_i u + c_i v), i = 1, 2, 3 \quad (15)$$

where A is the area of the triangular element, and a_i, b_i, c_i are coordinate transformation constants, which can be solved through the node coordinates. The more detailed derivation can be found in Reference [65]. Then, the strain function matrix can be derived as:

$$[B] = \frac{1}{2A} \begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix} \quad (16)$$

The expression for the element stiffness matrix of the wing membrane is:

$$K_w^e = \int_A B^T D B dA \quad (17)$$

where D is the elastic coefficient matrix of the wing membrane material.

The global stiffness matrix $[K_w]$ is assembled by the node order for each wing membrane region. In practical operations, to accelerate the computation, the global stiffness matrix can be defined as a sparse matrix. According to the stiffness equation, the in-plane load of its nodes is obtained as:

$$[P_w^{in}] = [K_w][\Delta r_w] \quad (18)$$

For the j -th wing membrane node, its in-plane load component on the Z_w axis of the wing coordinate system is:

$$P_{w,j}^{in,z} = P_{w,j}^{in} h_j \left(u_j^2 + v_j^2 + h_j^2 \right)^{-\frac{1}{2}} \quad (19)$$

The total load at each wing membrane node is obtained as:

$$P_{w,j} = P_{w,j}^{ext} + P_{w,j}^{in,z} \quad (20)$$

According to Assumption 2, the contact force distribution function at the wing vein boundary is:

$$R_j(u) = \begin{cases} \vartheta_j u + \varrho_j, 0 \leq u \leq u_j \\ \frac{\vartheta_j u_j}{u_j - u_L} (u - u_L) + \varrho_j, u_j < u \leq u_L \end{cases} \quad (21)$$

$$R_j(v) = \begin{cases} \varsigma_j v + \varrho_j, 0 \leq v \leq v_j \\ \frac{\varsigma_j v}{v_j - v_L} (v - v_L) + \varrho_j, v_j < v \leq v_L \end{cases} \quad (22)$$

where $\vartheta_j, \varrho_j, \varsigma_j$ are undetermined coefficients that can be solved through three equilibrium equations, namely the balance of vertical forces and the moments about two coordinate axes:

$$\begin{cases} \int_0^{u_L} R_j(u) du + \int_0^{v_L} R_j(v) dv = P_{w,j} \\ \int_0^{u_L} u R_j(u) du = P_{w,j} v_j \\ \int_0^{v_L} v R_j(v) dv = P_{w,j} u_j \end{cases} \quad (23)$$

Thus, the load at the i -th wing vein node can be expressed as:

$$P_{b,i} = P_{b,i}^{iner} + \sum R_j(r) l_e \quad (24)$$

where l_e represents the length of each beam element, and $P_{b,i}^{iner}$ represents the inertial force of the wing vein element under the action of flapping motion, can be expressed as:

$$P_{b,i}^{iner} = \rho_c \frac{\pi d_c^2}{4} l_e y_i^2 \frac{\partial \omega_\lambda}{\partial t} \quad (25)$$

where ρ_c is the density of the carbon fiber material, d_c is the diameter of the carbon fiber rod, and y_i is the spanwise coordinate of the midpoint of the beam element in the wing coordinate system.

The wing veins are treated as typical cantilever beam structures, with each node having two degrees of freedom, displacement and rotation. Its element stiffness matrix is [66]:

$$K_b^e = \frac{E_c I_c}{l_e^3} \begin{bmatrix} 12 & 6l_e & -12 & 6l_e \\ 6l_e & 4l_e^2 & -6l_e & 2l_e^2 \\ -12 & -6l_e & 12 & -6l_e \\ 6l_e & 2l_e^2 & -6l_e & 4l_e^2 \end{bmatrix} \quad (26)$$

where E_c is the elastic modulus of the carbon fiber, and I_c is its cross-sectional moment of inertia.

The global stiffness matrix $[K_b]$ for all wing veins is assembled by the node order of the element stiffness matrix K_b^e , and boundary conditions are applied using the penalty method. According to the stiffness equation, the deformation distribution matrix of each wing vein node is obtained as:

$$[q_b] = [K_b]^{-1} [P_b] \quad (27)$$

The normal displacement elements in $[q_b]$ are extracted by the degree of freedom order, and then the normal displacement array $[\Delta]$ of each wing membrane node in the Z_w axis direction is obtained by assigning values using a smooth Fourier function based on the mesh distribution and orientation.

2.3 Partitioned implicit algorithm

The computational results from the structural solver are ultimately fed into the QS model to update the relative AOA and the effective projected area of each element. The data exchange process employs a partitioned algorithm, which consists of multiple independent solvers that perform computations separately and then exchange data, significantly improving computational efficiency. To ensure the accuracy of the results, a partitioned implicit algorithm is used [67], involving multiple exchanges and iterative solving within each time step until the results converge to a predefined error limit. The overall data exchange process is illustrated in Fig. 3.

The overall methodology is as follows: Initially, the absolute AOA data from the fuselage is used as the initial calculation AOA for each wing membrane element. In the flow field solver, the QS model calculates the instantaneous normal force array $\{f_{inst}^e\}$ for each wing membrane element, which is then input into the structural field solver. Subsequently, the stiffness equation is solved to obtain the normal displacement array $[\Delta]$ for each wing node. By analyzing the dihedral angle relationship, the chord-wise twist angle array $\{\varepsilon_c^e\}$ and the spanwise twist angle array $\{\varepsilon_s^e\}$ for each wing element are derived. The $\{\varepsilon_c^e\}$ and $\{\varepsilon_s^e\}$ arrays can represent the inflow state distribution of the wing surface, where $\{\varepsilon_c^e\}$ updates the aerodynamic coefficients by modifying the relative AOA, and $\{\varepsilon_s^e\}$ reduces the effective projected area of

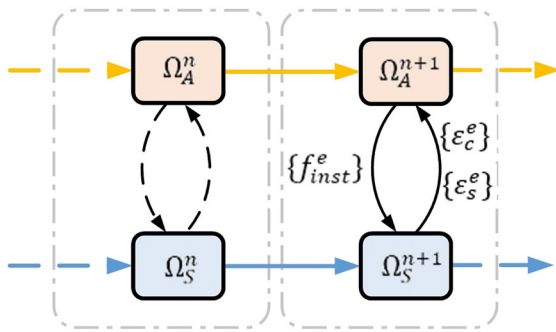


Fig. 3 Flowchart of the partitioned implicit algorithm. Ω_A^n represents the dynamic solver at the n -th discrete time step, and Ω_S^n represents the structural solver

the wing surface, thereby diminishing the effective aerodynamic forces. The updated inflow state is input back into the flow field solver to obtain a new set of instantaneous normal force arrays for the wing elements. This process repeats until the solvers exit the iterative loop and proceed to the next discrete time step.

Specifically, each wing node in the wing coordinate system has three coordinate dimensions: chordwise projection position, spanwise projection position, and normal displacement. The first two are determined and fixed during mesh generation, while the normal displacement changes with each iteration, consequently altering the node's three-dimensional coordinates. Since three-node shell elements are used as the basic elements for the wing surface, each wing element can be regarded as a plane defined by three nodes. The normal vector of this plane can be obtained via the vector cross product:

$$\bar{v}_n e = \bar{p}_1 e p_2^e \times \bar{p}_2 e p_3^e \quad (28)$$

where $p_i^e = (x_i, y_i)$ represents the coordinates of the counterclockwise ordered nodes of the triangular element.

As shown in Fig. 1d, the chordwise twist angle ε_c^e and the spanwise twist angle ε_s^e of the element can be derived using the dihedral angle formula:

$$\varepsilon_c^e = \frac{\pi}{2} - \arccos \left(\frac{\bar{v}_n^e \cdot \bar{g}_n}{|\bar{v}_n e| \cdot |\bar{g}_n|} \right) \quad (29)$$

$$\varepsilon_s^e = \frac{\pi}{2} - \arccos \left(\frac{\bar{v}_n^e \cdot \bar{b}_n}{|\bar{v}_n e| \cdot |\bar{b}_n|} \right) \quad (30)$$

where $\bar{g}_n$ is the normal vector of the $X_w O_w Z_w$ plane, and $\bar{b}_n$ is the normal vector of the $Y_w O_w Z_w$ plane.

The convergence criterion for the partitioned implicit algorithm is determined by calculating the difference in the normal displacement array $[\Delta]$ obtained from two consecutive operations. If the absolute value of each element in $[d\Delta]$ is less than the predefined error limit, the iteration loop exits, and the final results are output. Otherwise, bidirectional data exchange continues until the convergence criterion is met. In practice, a relaxation factor is introduced to control the displacement change range of the wing nodes in each computation step, keeping it less than 0.1 times the chord length [37] to prevent solver divergence.

3 Experimental validation of the proposed theory

In this section, we conducted force measurement experiments and motion posture monitoring experiments in a wind tunnel environment to obtain the mechanical data and flexible deformation data of FWAVs under constant-speed cruising conditions. The FWAVs prototypes were successfully manufactured and equipped with various wings of different spans, chord lengths, and wing vein layouts. During the entire experiment, the flight AOA, constant cruising speed, and flapping frequency could be accurately adjusted. By comparing the experimental data with theoretical estimates and analyzing the errors, we determined the effective range of the proposed theory.

3.1 Experimental platform

The wind tunnel experiments were conducted at the Wind Tunnel and Water Flume of Harbin Institute of Technology (WTWF-HIT). The type of wind tunnel used was a single-return closed-loop wind tunnel, with an error margin of less than 0.1 m/s for wind speeds below 10 m/s. The test section dimensions are 4 m $\times$ 3 m $\times$ 25 m, which sufficiently meets the requirement of reducing near-wall boundary effects

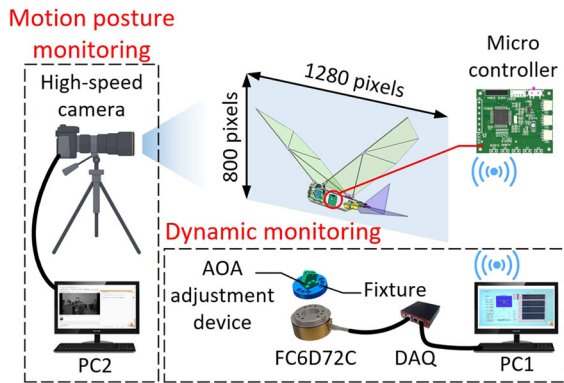


Fig. 4 Hardware system for wind tunnel experiments. It includes motion posture monitoring system, dynamics testing system and remote control sensing system

for bird-sized FWAVs with wingspans of less than 1 m.

The hardware system configuration is shown in Fig. 4, and the physical photos in the wind tunnel experimental site is shown in Fig. 5. The flapping motion mechanism is a common spatial four-bar linkage mechanism [55], and the flapping frequency is monitored and controlled in a closed-loop manner using a Hall sensor in the secondary reducer, with a steady-state error of less than 0.1 Hz. The speed control commands are transmitted and received wirelessly. The AOA adjustment device is a series of pre-designed 3D-printed components with preset installation AOAs. It connects the six-axis force sensor to the FWAVs prototype and allows for manual disassembly and assembly to achieve different installation angles, with an error margin of less than 0.5° . Motion posture detection experiments are conducted using a high-speed camera (VEO-E 310L, Vision Research, Inc.) with a shooting frequency of up to 1000 Hz. Due to layout restrictions in the wind tunnel, the high-speed camera was positioned at a single viewpoint directly in front of the FWAVs prototype. By tracking marked points, all necessary deformation data can be obtained. Force measurement experiments use a six-axis force/torque sensor (FC6D72C, ForceChina, Inc.) with a sampling frequency of up to 1000 Hz and a force resolution of 1/20 N. A fifth-order low-pass Butterworth filter is applied to the force/torque data, with a cutoff frequency set to five times the flapping frequency to filter out high-frequency vibrations while retaining reliable force data.

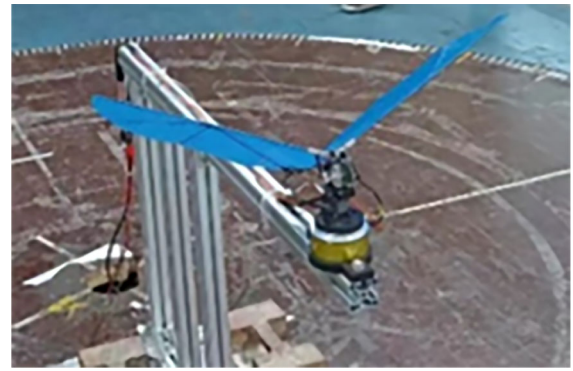


Fig. 5 Physical photos in the wind tunnel experimental site. The FWAVs prototype and testing devices are mounted on a 1.5 m-high aluminum frame

3.2 Motion posture monitoring experiment

In previous studies, to accurately describe the flexible deformation of wings during flapping, numerous markers are usually introduced for high-precision motion capture [39], or dynamic 3D reconstruction using depth cameras is directly adopted [38]. However, these approaches are typically due to specific requirements, such as establishing a data-driven dynamic model [39]. In this study, the motion posture detection experiment aims solely to validate the accuracy of the structural solver. Therefore, marking only 4–5 key markers on one side of the symmetric wings is sufficient, focusing solely on their displacement changes in the Z_w axis direction in the wing coordinate system. Figure 6 shows a frame captured by the high-speed camera, where the origin O_w of the wing coordinate system is the rotation center of the wing leading edge axis, with fixed coordinates. Point P_0 is the connection point at the wing root, aiding in calibrating the flapping angle θ at the current time frame. Point P_1 is the distal end of the wing leading edge axis, while points P_2 and P_3 are the distal ends of the wing veins experiencing significant deformation.

The pixel coordinates of P_1, P_2, P_3 are recorded and then transformed into the wing coordinate system. The coordinate transformation equation from the camera coordinate system to the wing coordinate system is:

$$p_w = k_{cw} R_{cw} (p_c + T_{cw}) \quad (31)$$

where p_w and p_c represent the coordinates of the marker points in the wing coordinate system and the camera coordinate system, respectively. k_{cw} is the

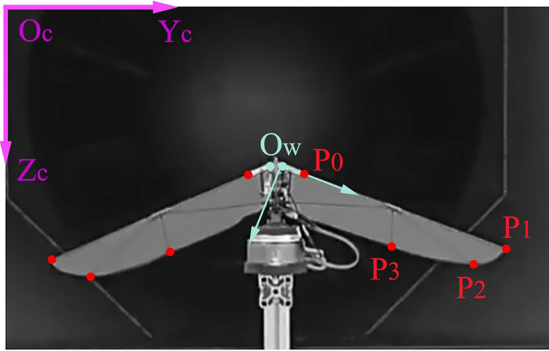


Fig. 6 Markers in the motion posture monitoring experiment. $Y_c O_c Z_c$ represents the fixed camera coordinate system, while $Y_w O_w Z_w$ represents the wing coordinate system, which varies over time

projection and scaling matrix, R_{cw} is the rotation matrix, and T_{cw} is the translation matrix, expressed as:

$$k_{cw} = \text{diag} \left\{ \frac{k_\lambda}{\cos \alpha_0}, \frac{k_\lambda}{\cos \alpha_0} \right\} \quad (32)$$

$$R_{cw} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (33)$$

$$T_{cw} = (\overrightarrow{O_c O_w}) \quad (34)$$

where k_λ is the global scaling coefficient, obtained by measuring the pixel vector length between point O_w and marker point P_1 in the static state of the wing with an initial flapping angle of 0 degrees and converting it to the actual wingspan scale.

To validate the accuracy and broad applicability of the improved QS model proposed in this paper, numerous comparative experiments were conducted under different flapping frequencies, different cruising flight speeds, different body absolute angles of attack, and even different wing shapes. The constant parameters of the wing used in the experiment were as follows: a single-side wingspan of 300 mm, a chord length of 160 mm, an amplitude of 70° , with the wing membrane made of P31N parachute fabric and the wing veins made of carbon fiber rods. When computing model data, the finite element mesh consisted of 22,500 elements, and the experimental data passed a grid-independence test, showing less than 1% change in results when the mesh count was doubled. Figure 7 shows the time history of normal deformation data of marker points P_1, P_2, P_3 in the wing coordinate

system, which including the combined effects of wing bending and twisting. The solid lines represent the predicted data from the model, while the scattered points represent the test data from the wind tunnel experiments.

Figure 7a–c illustrate the effect of flapping frequency on wing flexible deformation. As the flapping frequency increases, the aerodynamic force increases, leading to greater flexible deformation, without significant attenuation below 8 Hz. When the flapping frequency exceeds 10 Hz, the experimental data gradually stabilizes as the flexible deformation approaches the structural limit. However, the model's predicted data shows that the flexible deformation continues to increase as the flapping frequency rises. The prediction error of the model reaches its peak around the mid-cycle of the flapping motion, when the downward wingbeat velocity is at its maximum. This error is caused by the assumption of small deformation. However, bird-sized FWAVs typically do not operate at high flapping frequencies, and the model prediction accuracy at lower frequencies is sufficient.

Figure 7d–f show the effect of body absolute AOA on wing flexibility deformation under constant-speed cruising. With a constant flapping frequency and incoming flow speed, a small range change in the AOA does not significantly affect wing flexibility deformation, but an increase in positive AOA causes a shift in overall deformation in the positive direction. This indicates that rotational circulation force is almost unaffected by body AOA, while the translational circulation force positively correlates with body AOA within a small range.

Figure 7g–i illustrate the effect of incoming flow speed on wing flexibility deformation. As the incoming flow speed increases, the wing flexibility deformation is slightly suppressed, with the most significant suppression near the whole cycle (maximum upward flapping speed). This is because, at a positive AOA, the upward flapping motion of the wing generates a chordwise torsion angle in the opposite direction to that induced by the incoming flow. Around mid-cycle (when the wing's downward flapping speed is at its peak), the wing's downward motion produces a positive chordwise torsion angle, reducing the value of the relative AOA on each element. As a result, the influence of incoming flow speed on the aerodynamic characteristics is diminished, and the impact of wind

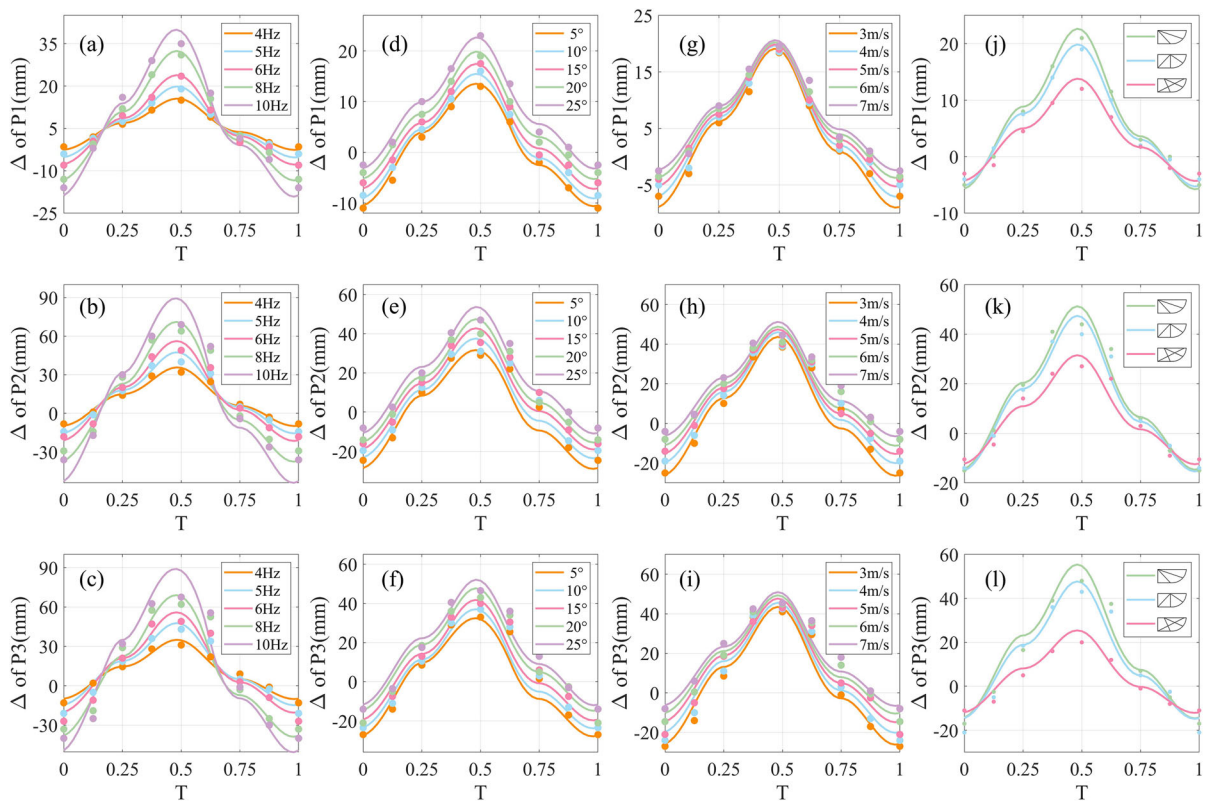


Fig. 7 Time history of flexible deformation over a flapping cycle. The first row corresponds to the deformation data of marked point P_1 , the second row corresponds to P_2 , and the third row to P_3 . The first column contains data tested under different flapping frequencies, with a 20-degree AOA, a wind speed of

5 m/s, and using the second veins layout configuration. The remaining columns use a fixed flapping frequency of 5 Hz. The second column's variable is AOA, the third column's variable is the incoming flow velocity, and the fourth column's variable is the veins layout configuration

speed changes on the wing's flexible deformation is also reduced.

Figure 7j–l illustrate the effect of different wing stiffness distributions on wing flexibility deformation. The three wing shapes shown have the same wing vein distal endpoints, resulting in identical initial positions of the marker points. Wing type 1 has the lowest spanwise and chordwise stiffness, while wing type 3 has the highest. The deformation of point P_1 reflects the spanwise stiffness, while points P_2 and P_3 reflect the chordwise stiffness. Greater wing flexibility results in greater flexible deformation, especially near the half-cycle, because during the downward flapping motion at a positive angle of attack, the direction of the chordwise torsion angle generated by the wing aligns with that induced by the incoming flow. This alignment leads to peak aerodynamic force and significant deformation.

The main sources of error are manufacturing inaccuracies, such as dimensional precision and installation gaps. Particularly, the membrane's flatness between the wing veins cannot be guaranteed, with small wrinkles reducing in-plane tensile stress and diminishing its suppression of normal flexible deformation. Additionally, the detachable wing structure causes gaps and rotational issues around the leading-edge axis, affecting points P_2 and P_3 more significantly, especially during downward acceleration (around $t = 0.375T$) or deceleration (around $t = 0.625T$). Nonetheless, the improved QS model proposed in this paper still exhibits high prediction accuracy for wing flexibility deformation and broad applicability, reflecting the impact of various design parameters on aerodynamic characteristics. The root mean square error (RMSE) for each set of flight parameters over one flapping cycle was calculated, and the percentage root mean square error (PRMSE)

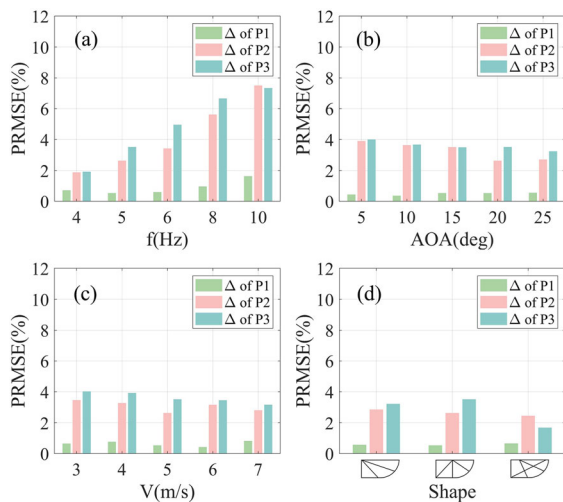


Fig. 8 Percentage root mean square error of flexible deformation over a flapping cycle. **a** Accuracy at different flapping frequencies. **b** Accuracy at different AOA. **c** Accuracy at different incoming flow velocities. **d** Accuracy at different wing vein layouts. The three colors represent the tracking errors for the three marker points, respectively

was defined by normalizing RMSE to the wing chord length. The results are summarized in Fig. 8, showing that the overall error remains within 8%. Furthermore, in conditions closer to actual flight scenarios with non-high-frequency flapping, the error can be reduced to below 5%, demonstrating excellent computational accuracy.

3.3 Dynamic monitoring experiment

When analyzing the dynamic characteristics of FWAVs, we focus more on the mean data over several stable flapping cycles rather than the large, noisy time history data, as mean data intuitively indicate the flying capability of the designed vehicle. The dynamic data of FWAVs were directly measured using a six-axis force sensor. In the wind tunnel experiment, the trajectory coordinate system and the six-axis force sensor coordinate system overlap due to the constant-speed cruising assumption. Only the forces in the lift and thrust directions are considered, and the mean data are calculated over five complete flapping cycles. Numerous comparative experiments were conducted under different flapping frequencies, body installation angles, incoming flow speeds, and wing shapes. Control groups without wings were used to eliminate the influence of the fuselage and tail on aerodynamic

characteristics. The processed wing dynamic data are summarized in Fig. 9, including wind tunnel measured force data, predictions from the improved QS model proposed in this paper, and predictions from the traditional QS model.

Here, the traditional QS model is based on BEM, assuming that the distribution of wing flexibility is regular, allowing the pitch function of the wing to be estimated directly through empirical formulas without any iterative calculations. Let ε_{cm} represent the average amplitude of the chordwise torsion angle measured from experimental data, then the approximate expression of the chordwise torsion angle is given as [31]:

$$\varepsilon_c = \varepsilon_{cm} \sin\left(2\pi ft + \frac{\pi}{2}\right) \quad (35)$$

The mean absolute errors (MAE) of lift and thrust for the two models were then calculated, and the percentage mean absolute error (PMAE), defined as the ratio of MAE to the prototype's weight, was summarized in Fig. 10.

Figure 9a and b show the impact of different flapping frequencies on aerodynamic characteristics. The improved QS model demonstrates high prediction accuracy for aerodynamic characteristics at frequencies below 8 Hz, with significant errors only arising when the flapping frequency exceeds 9 Hz due to the small deformation assumption. As shown more clearly in Fig. 10a, the modeling method proposed in this study maintains relative errors below 5% for frequencies under 9 Hz. In contrast, the traditional QS model, which estimates the chordwise torsion angle using empirical formulas, fails to accurately track changes in aerodynamic characteristics as the flapping frequency increases. Its prediction accuracy is only reliable near 5 Hz, where the empirical formulas were derived from experimental data. Once the flapping frequency deviates from this range, the prediction error increases rapidly.

Figure 9c and d display the influence of different absolute angles of attack on aerodynamic characteristics. Since the empirical formula of the QS model is a function of the AOA, both the traditional and improved QS models exhibit good prediction accuracy. As shown in Fig. 10b, when the AOA reaches the relatively high value of 25°, the traditional QS model, unable to account for actual flexible deformation, shows a slightly larger error, reaching around 7%. In

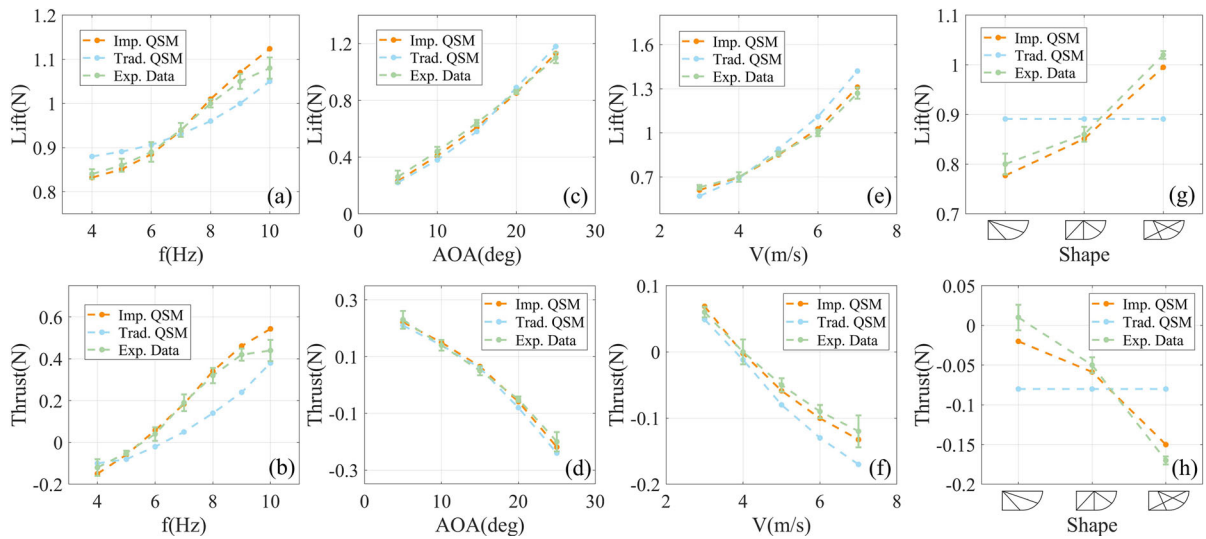


Fig. 9 Mean aerodynamic force data over different parameter groups. The definitions of each column for the tested variables are consistent with Fig. 7. The first row shows the average lift data, and the second row shows the average thrust data. The

three colors respectively represent the calculated data from the improved QS model, the traditional QS model, and the force measurement experiment

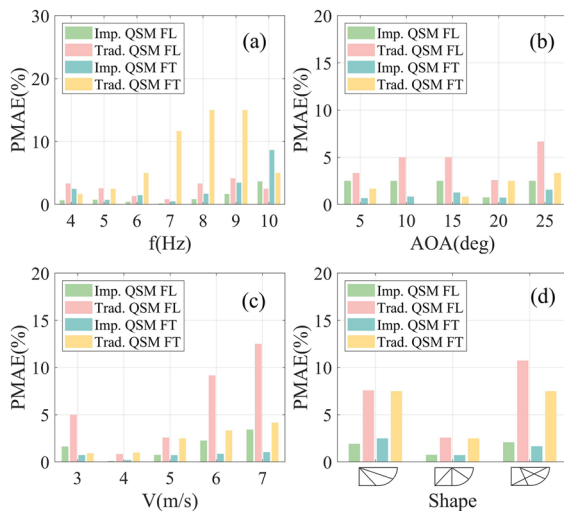


Fig. 10 Percentage mean absolute errors over different parameter groups. The definition of the subplots is consistent with Fig. 8. The four colors represent the following, in order: the lift error of the improved QS model, the lift error of the traditional QS model, the thrust error of the improved QS model, and the thrust error of the traditional QS model

contrast, the global prediction error of the improved QS model remains below 2.5%, demonstrating excellent accuracy.

Figure 9e and f illustrate the effect of different incoming flow speeds on aerodynamic characteristics. As discussed in the explanation of Fig. 7g–i, the

relationship between incoming flow speed and flexible deformation has been clearly established. The improved QS model accurately predicts this relationship, and as shown in Fig. 10c, it maintains good prediction accuracy even under high wind speed conditions, with relative errors consistently below 5%. In contrast, the traditional QS model, which cannot accurately estimate the variation in flexible deformation with wind speed, experiences a sharp increase in error, reaching nearly 15% under high wind speed conditions.

Figure 9g and h depict the impact of different wing stiffness distributions on aerodynamic characteristics. The greater the wing stiffness, the smaller the flexible deformation, reflected as an increase in lift and a decrease in thrust. In the actual design process of FWAVs, the impact of wing flexibility must be fully considered. If the flexibility is too large, sufficient lift cannot be obtained, whereas if the flexibility is too small, sufficient thrust may not be generated. Appropriate flexibility is needed to achieve better aerodynamic characteristics. The improved QS model can accurately predict this relationship, whereas the traditional QS model, driven by empirical formulas, cannot quantitatively calculate wing flexible deformation and, therefore, cannot distinguish the aerodynamic characteristics of different wing vein layouts.

In summary, since the bird-sized FWAVs we are studying do not operate in high-frequency flapping states during real flight operations, the conclusion is that the global prediction errors for lift and thrust using the improved QS model remain below 5%. In comparison, the traditional QS model, which cannot quantitatively describe flexible deformation, only maintains low error within a narrow range where its empirical coefficients are applicable. However, under high-frequency and high wind speed conditions, its error can exceed 15%. Furthermore, the improved QS model can analyze the impact of different wing vein layouts on aerodynamic characteristics, which the traditional QS model cannot achieve. This high-precision, improved QS model has significant application prospects in guiding the structural and controller design of FWAVs.

To highlight the balance between computational efficiency and accuracy in the proposed model, this study compares it with the CFD method. Given the significant computational resources required by the CFD approach, extensive data comparisons, as done with the traditional QS model, are not feasible. Therefore, a single computation result is shown as an example under the conditions of wing type 2, 5 Hz flapping frequency, 5 m/s wind speed, and an installation AOA of 20 degrees. The CFD method is implemented through a co-simulation using the commercial software XFlow and Abaqus, where XFlow calculates the flow field and Abaqus handles the structural deformation. A built-in interface facilitates data exchange, enabling bidirectional fluid–structure interaction analysis. The virtual wind tunnel has dimensions of $2\text{ m} \times 2\text{ m} \times 4\text{ m}$, with a global resolved scale of 0.01 m. An adaptive refinement algorithm further refines the grid resolution in the vicinity of the wing and wake regions to 0.0025 m. In Abaqus, the wing membrane is modeled as a shell structure, while the veins are modeled as beam structures, with binding constraints applied at the nodes. The flapping motion is generated by applying angular displacement boundary conditions, with each flapping stroke discretized into 100 time steps. To ensure that the periodic wake generated by the flapping motion reaches stability, a total of 8 flapping strokes are computed, with the average data from the last 3 strokes used as the CFD results. The summarized data is shown in Table 1. It can be seen that, when using a comparable grid resolution for the wing

Table 1 Comparison of the proposed modeling method and the CFD method in terms of accuracy and efficiency

Schemes adopted	Improved QS model	CFD method
Average lift data	0.851 N	0.856 N
PMAE of lift	0.75%	0.33%
Average thrust data	−0.0587 N	−0.0612 N
PMAE of thrust	0.73%	0.93%
CPU-time (i7-10875H)	49.6 s	64.2 h

surface, the proposed improved QS model achieves similar accuracy to the CFD method while consuming 3–4 orders of magnitude less time, demonstrating an effective trade-off between accuracy and efficiency.

4 Application of the proposed theory

Due to material limitations, the FWAVs' prototype structures, movement patterns, and sensing systems cannot fully mimic real birds, resulting in much lower flight efficiency compared to real birds. The structural parameters designed using the bionic scaling law [68] are often unsuitable. Generally, the structural design of FWAVs requires extensive experimental data to support it. By comparing and analyzing the effects of different angles of attack, incoming flow speeds, flapping frequencies, wing areas, and aspect ratios on aerodynamic characteristics, a suitable set of flight parameters can gradually be found. However, the

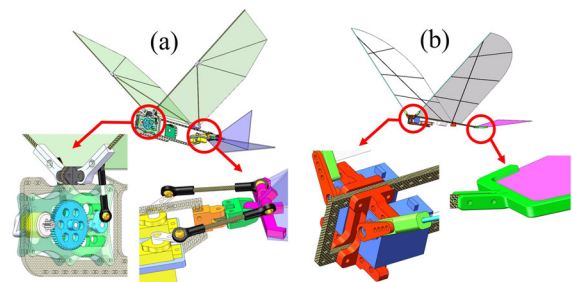


Fig. 11 3D design models of two FWAV prototypes. **a** Prototype 1: Uses a spatial four-bar linkage mechanism to convert the continuous rotational motion from the motor into flapping wing motion. **b** Prototype 2: Achieves flapping motion by continuously switching the servo drive unit between forward and reverse rotations

monetary cost of wind tunnel experiments is prohibitively high, and the time cost of CFD simulation methods is also extremely high. The improved QS model proposed in this study balances low computational cost with high accuracy. Based on this improved QS model, we have developed various bird-mimicking FWAVs based on continuous rotation mechanisms and micro servo-driven mechanisms, achieving flapping flight with open-loop control at various flapping frequencies and wingspans, fully demonstrating the model's application value.

These two FWAVs are designed for different operational scenarios (Fig. 11). Prototype 1 has excellent power output capabilities, can carry a higher load, and has a longer endurance. Prototype 2 can fully control wing angular position and angular velocity, achieving three-axis attitude control with just two servo motors [63, 64, 69]. It has better bionic effects, lighter weight, and better maneuverability. This means Prototype 1 has a higher body weight and requires greater aerodynamic lift but can flap its wings faster, while Prototype 2 is the opposite. The improved QS model can simulate the aerodynamic characteristics of both design targets well, aiding in the structural parameter design tasks.

The specific design steps can be divided into the following phases:

Step 1: Propose Initial Design Parameters. We aim to design a medium-sized bird-mimicking FWAVs (with a wingspan between 0.5 m and 1 m). This size balances load-carrying capacity and flexibility, capable of mounting various small sensors like IMUs, GPS, barometers, and optical flow modules. The smaller turning radius reduces the requirements for the test site. Using 3D modeling software for mass property assessment, the approximate weight of the prototypes carrying various sensors and communication modules is estimated to be around 120 g (for Prototype 1) and 80 g (for Prototype 2). The wing's aspect ratio is also one of the initial design parameters. In Sect. 3, we did not study the effect of aspect ratio on aerodynamic characteristics because this parameter needs to consider many non-aerodynamic factors in actual design tasks. Generally, a larger aspect ratio improves aerodynamic characteristics but usually comes with poor passability and flexibility issues. More critically, increasing the aspect ratio also increases the body's rotational inertia and aerodynamic moment arm, requiring more control and

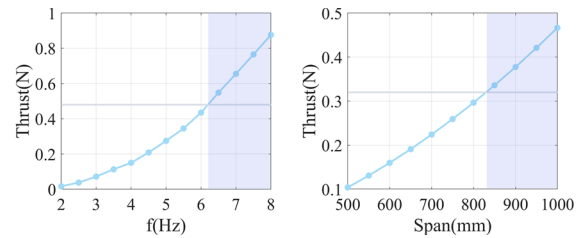


Fig. 12 Aerodynamic characteristics with varying pending design parameters. **a** The design parameter to be determined is the flapping frequency. **b** The design parameter to be determined is the wingspan length

driving torque. Considering all factors, an aspect ratio of 4–6 is acceptable in practical design tasks. In this case, the value is set to 4.

Step 2: Determine Pending Design Parameters. For Prototype 1, the gearbox reduction ratio can be freely configured, meaning the upper limit of the flapping frequency is adjustable. Therefore, the flapping frequency is set as a pending design parameter, while the wing span is set as the initial design parameter, which is 0.6 m in this case. Prototype 2 is limited by the output capacity of commercially available micro servo motors, with a maximum continuous bidirectional rotation frequency of about 4 Hz. Therefore, the wingspan is set as the pending design parameter.

Step 3: Calculate Pending Design Parameters. Based on our research experience, the thrust-to-weight ratio for small to medium-sized flat wing FWAVs with a wingspan in the range of 0.5 m to 1 m needs to reach at least 0.4 during the takeoff phase to ensure successful lift-off and a certain level of wind resistance. The proposed improved QS model can accurately calculate the thrust even in zero wind speed and zero AOA conditions, ensuring high accuracy in the calculation of aerodynamic characteristics. Figure 12 shows the aerodynamic characteristics as the pending design parameters gradually increase, with the shaded area representing the feasible range where the thrust-to-weight ratio is greater than 0.4. As shown in Fig. 12a, Prototype 1 needs to achieve a flapping frequency of at least 6.2 Hz. As shown in Fig. 12b, Prototype 2 needs a wingspan of at least 830 mm.

Step 4: Validate Feasible Flight Parameter Space. FWAVs prototypes need to meet two necessary conditions during takeoff and climb: (1) lift greater than weight, and (2) thrust greater than zero. Only increasing flapping frequency or wing area can

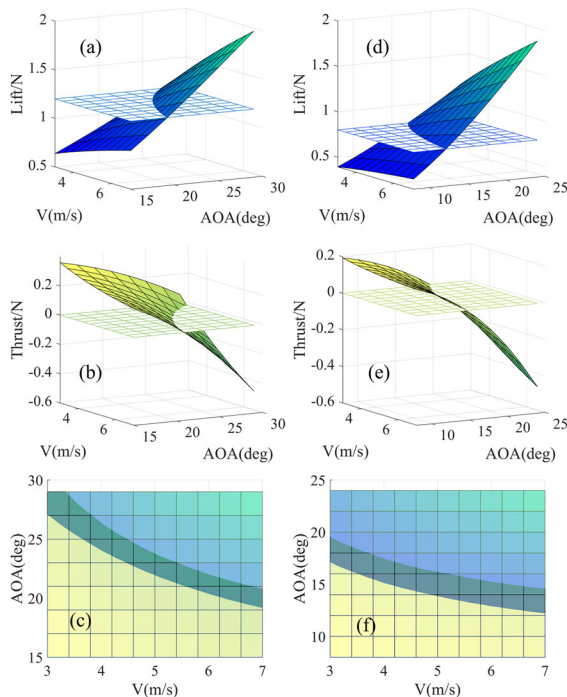


Fig. 13 Verification of Feasible Flight Parameter Space. The first column corresponds to the validation of Prototype 1, and the second column corresponds to the validation of Prototype 2. The first row is used to identify the parameter range that meets the lift conditions, and the second row is used to identify the parameter range that meets the thrust conditions. The third row integrates the data from the first and second rows to display the overlapping parameter range that satisfies both lift and thrust requirements

simultaneously increase lift and thrust. Changing the AOA or flight speed can only increase one while decreasing the other. Therefore, it is necessary to verify whether the designed structural parameters have a feasible range in the flight parameter space. By traversing the AOA and flight speed within common parameter ranges, the flight parameter space is obtained as shown in Fig. 13. Figure 13a–c show the model data for Prototype 1, and Fig. 13d–f show the model data for Prototype 2. Figure 13a–d show the aerodynamic lift surface and prototype weight plane, with the part of the surface above the plane enabling the FWAVs to climb. Figure 13b–e show the aerodynamic thrust surface and zero drag plane, with the part of the surface above the plane enabling the FWAVs to accelerate forward. The overlapping area in the top view of the extracted Fig. 13c–f indicates the feasible region. The width of about 2–3° in the AOA axis is considered reasonable. If the overlapping area is too



Fig. 14 Verification of Indoor and Outdoor Flight Capabilities. **a** shows Prototype 1 flying in an indoor corridor, **b** shows Prototype 1 flying in an open area, **c** shows Prototype 2 flying in a forest, and **d** shows Prototype 2 flying between buildings

small or non-existent, it means the FWAVs prototype has strict flight environment requirements and is challenging to control. If the overlapping area is too large, it means the designed driving performance is excessive and may not match suitable driving components.

Step 5: Selection, Design, and Verification of Key Components. Choose the drive motor model for Prototype 1 as EMAX-RS1108, and the designed two-stage gearbox reduction ratio is about 61. The servo motor model for Prototype 2 is GDW-DS290MG.

Step 6: Flight Experiment Verification. Flight experiments of the two designed FWAVs prototypes were conducted in various environments (Fig. 14). All flight experiments were conducted using open-loop control. These experiments demonstrate that the flapping-wing prototype, designed based on the modeling method proposed in this study, generates sufficient lift and thrust to support flight operations, thereby validating the effectiveness of the proposed model.

5 Conclusions

This study proposes an improved QS model capable of quantitatively calculating the flexible deformation of wings, reducing the dependence of the dynamic model on empirical coefficients and achieving a favorable balance between computational efficiency and accuracy. The proposed theory has been validated through motion attitude monitoring experiments and dynamic monitoring experiments. Within the common flight

parameter range, the model's relative calculation error for aerodynamic forces is within 5%. Additionally, based on the proposed theory, two FWAV prototypes with different drive mechanisms were designed. The effectiveness of the designed structures and the reliability of the design methods were validated through indoor and outdoor flight experiments. In the future, we will continue to explore the application of the improved QS model in further optimizing structural design and guiding controller design, aiming to achieve more stable and agile flight.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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